

# fine\_tune\_BERT\_keras

November 22, 2025

## 1 66% Balanced Consensus: Fine-tuning BERT as a Financial Sentiment Classifier (Keras)

### 1.1 30 epochs with model checkpointing based on validation F1

This document provides an explanation of the python script used to train a classifier on top of a BERT model via fine-tuning. The task is financial sentiment analysis, where sentences are classified into three sentiments: negative, neutral, and positive.

### 1.2 Dataset

The dataset is [hosted on Hugging Face](#), where you can find further information on its curation, motivation, and annotation.

#### 1.2.1 Overview

The dataset consists of 4846 lines, each containing a financial text sentence and its corresponding sentiment, separated by an '@' symbol. The sentiment can belong to one of three categories: Negative, Neutral, or Positive, and is distributed as:

- Neutral: 2879 entries (59%)
- Positive: 1363 entries (28%)
- Negative: 604 entries (12.4%)

#### 1.2.2 Data Fields

- **sentence:** A tokenized line from the dataset.
- **label:** A label representing the sentence class as a string: 'positive', 'negative', or 'neutral'.

#### 1.2.3 Data Splits

Although the dataset does not come with predefined training/validation/test splits, the sentences are categorized into different configurations based on the degree of agreement among the annotators:

- **sentences\_50agree:** Number of instances with  $\geq 50\%$  annotator agreement - 4846 sentences
- **sentences\_66agree:** Number of instances with  $\geq 66\%$  annotator agreement - 4217 sentences
- **sentences\_75agree:** Number of instances with  $\geq 75\%$  annotator agreement - 3453 sentences
- **sentences\_allagree:** Number of instances with 100% annotator agreement - 2264 sentences

For this project, we use the `sentences_50agree` configuration comprising of 4846 instances and manually split it into training, validation, and testing sets in a 70-15-15 proportion.

All sentences in the dataset are limited to fewer than 512 tokens, in accordance with the BERT model's input constraints.

### 1.3 Imports and System Setup

The first part of the code includes importing necessary modules and setting up the TensorFlow environment to recognize accessible GPU devices.

### 1.4 Metrics

The model's performance after training is evaluated based on accuracy and F1 score. Accuracy measures the proportion of correct predictions, while the F1 score provides a harmonic mean between precision and recall, hammered down for individual classes.

F1 score is calculated after each epoch by a custom `F1_Evaluation` class that uses the Scikit-learn function `f1_score()` for calculation.

We took the extra step of tracking the F1 score for the validation set for a few reasons due to using a financial dataset:

- **Handling Imbalance:** Financial datasets often exhibit class imbalance. For instance, genuine transactions significantly outnumber fraudulent ones. Accuracy can be misleading in such cases, making F1 a more balanced metric as it considers both precision and recall.
- **Balancing Costs:** Both false negatives (e.g., overlooking a negative sentiment) and false positives (e.g., misclassifying a neutral statement as negative) have implications. The F1 score ensures a balance between reducing these two error types.
- **Threshold Tuning:** Financial decisions can involve adjustable thresholds. The F1 score can guide optimization, ensuring neither precision nor recall is neglected.

### 1.5 Wandb

[Wandb \(Weights & Biases\)](#) is used for experiment tracking.

### 1.6 Preprocessing

The data is read from the `.txt` file. The file encoding is first detected using the `chardet` library, and then the data is loaded into a pandas `DataFrame` using the detected encoding. The `DataFrame` contains two columns: `sentence` and `label`.

Then the data is split into train, validation, and test sets, stratifying on the label to respect the original distribution of classes in the dataset.

The `BertTokenizer` is used to encode the sentences into format suitable for the BERT model. `truncation=True` and `padding=True` ensures that all sequences are of the same length.

The tokenized sentence encodings and the corresponding labels are wrapped into TensorFlow's `Dataset` objects.

## 1.7 Model Creation

The existing BERT model ('bert-base-uncased') from HuggingFace's transformer models is loaded for sequence classification with 3 labels (negative, neutral, positive).

## 1.8 Model Compilation and Training

We compile the model with the Adam optimizer, SparseCategoricalCrossentropy loss, and accuracy as a metric. We use SparseCategoricalCrossentropy because this is a multi-class classification problem with three integer labels (negative, neutral, positive), not one-hot vectors. Concretely, for each input example the model outputs a length-3 vector of raw scores called logits, one score per class. The loss function then applies a softmax to those logits to turn them into a probability distribution over the three classes.

Given a true label  $y \in 0, 1, 2$  (encoded as a single integer, not a one-hot vector), the loss for that example is the negative log of the predicted probability assigned to class  $y$ . At inference time, we take the argmax of the logits (or of the softmax probabilities) to pick the predicted class: the index of the largest score corresponds to "negative", "neutral", or "positive" respectively.

A Wandb run is initialized, and the data is fit on the model using a batch size of 8 for 30 epochs. The validation data and the custom F1 score calculation callback are also passed in the fit method, as is early stopping.

## 1.9 Evaluation

The model is evaluated on the test set. Classification report and F1 score are printed to show detailed information about performance on all three categories. The F1 score is also logged to the Wandb run.

```
[43]: # Installation printed output removed
!pip install wandb
```

```
Requirement already satisfied: wandb in /usr/local/lib/python3.12/dist-packages
(0.23.0)
Requirement already satisfied: click>=8.0.1 in /usr/local/lib/python3.12/dist-
packages (from wandb) (8.3.1)
Requirement already satisfied: gitpython!=3.1.29,>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from wandb) (3.1.45)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-
packages (from wandb) (25.0)
Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-
packages (from wandb) (4.5.0)
Requirement already satisfied: protobuf!=4.21.0,!5.28.0,<7,>=3.19.0 in
/usr/local/lib/python3.12/dist-packages (from wandb) (5.29.5)
Requirement already satisfied: pydantic<3 in /usr/local/lib/python3.12/dist-
packages (from wandb) (2.11.10)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from wandb) (6.0.3)
Requirement already satisfied: requests<3,>=2.0.0 in
/usr/local/lib/python3.12/dist-packages (from wandb) (2.32.4)
```

Requirement already satisfied: sentry-sdk>=2.0.0 in  
 /usr/local/lib/python3.12/dist-packages (from wandb) (2.45.0)  
 Requirement already satisfied: typing-extensions<5,>=4.8 in  
 /usr/local/lib/python3.12/dist-packages (from wandb) (4.15.0)  
 Requirement already satisfied: gitdb<5,>=4.0.1 in  
 /usr/local/lib/python3.12/dist-packages (from gitpython!=3.1.29,>=1.0.0->wandb)  
 (4.0.12)  
 Requirement already satisfied: annotated-types>=0.6.0 in  
 /usr/local/lib/python3.12/dist-packages (from pydantic<3->wandb) (0.7.0)  
 Requirement already satisfied: pydantic-core==2.33.2 in  
 /usr/local/lib/python3.12/dist-packages (from pydantic<3->wandb) (2.33.2)  
 Requirement already satisfied: typing-inspection>=0.4.0 in  
 /usr/local/lib/python3.12/dist-packages (from pydantic<3->wandb) (0.4.2)  
 Requirement already satisfied: charset\_normalizer<4,>=2 in  
 /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.0.0->wandb) (3.4.4)  
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-  
 packages (from requests<3,>=2.0.0->wandb) (3.11)  
 Requirement already satisfied: urllib3<3,>=1.21.1 in  
 /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.0.0->wandb) (2.5.0)  
 Requirement already satisfied: certifi>=2017.4.17 in  
 /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.0.0->wandb)  
 (2025.11.12)  
 Requirement already satisfied: smmap<6,>=3.0.1 in  
 /usr/local/lib/python3.12/dist-packages (from  
 gitdb<5,>=4.0.1->gitpython!=3.1.29,>=1.0.0->wandb) (5.0.2)

```
[44]: # Installation printed output removed
!pip install transformers
```

Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-  
 packages (4.57.1)  
 Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-  
 packages (from transformers) (3.20.0)  
 Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in  
 /usr/local/lib/python3.12/dist-packages (from transformers) (0.36.0)  
 Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-  
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 Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-  
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 Requirement already satisfied: regex!=2019.12.17 in  
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 Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-  
 packages (from transformers) (2.32.4)  
 Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in  
 /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.1)  
 Requirement already satisfied: safetensors>=0.4.3 in

```

/usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-
packages (from transformers) (4.67.1)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.34.0->transformers) (2025.3.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.34.0->transformers) (4.15.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.34.0->transformers) (1.2.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->transformers) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers)
(2025.11.12)

```

```

[45]: # Easy place to track what we're training
# Use the 66% consensus dataset
WHICH_DATASET = 'Sentences_66Agree.txt'
# Save our model checkpoints to
SAVE_PATH = "/content/drive/My Drive/finetuned_bert_66_30"

# Define the Weights & Biases run name
WHICH_RUN_NAME = f"bert_finetune_66_agree_{WHICH_DATASET.split('.')[0]}"

seed_val = 42

# Import necessary libraries
import os

# Used to suppress output for F1 validation calculation
import io
import contextlib

# For encoding detection
import chardet

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

```

```

from sklearn.metrics import classification_report, f1_score
from sklearn.model_selection import train_test_split

# Hugging Face Transformers for BERT
from transformers import BertTokenizer, TFBertForSequenceClassification

# Deep learning frameworks
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import backend as K
from tf_keras.callbacks import EarlyStopping

# Ensure we have a GPU
print("Num GPUs Available: ", len(tf.config.list_physical_devices('GPU')))

# Verify current GPU
device_name = tf.test.gpu_device_name()
if device_name != '/device:GPU:0':
    raise SystemError('GPU device not found')
print('Found GPU at: {}'.format(device_name))

# Weights and Biases for tracking experiments
import wandb
from wandb.integration.keras import WandbCallback

from google.colab import drive
drive.mount('/content/drive')

```

```

Num GPUs Available:  1
Found GPU at: /device:GPU:0
Drive already mounted at /content/drive; to attempt to forcibly remount, call
drive.mount("/content/drive", force_remount=True).

```

```

[46]: print(type(early_stopping))
      print(type(f1_evaluation))
      print(type(WandbCallback(save_model=False)))

```

```

<class 'tf_keras.src.callbacks.EarlyStopping'>
<class '__main__.F1_Evaluation'>
<class 'wandb.integration.keras.keras.WandbCallback'>

```

## 2 Prepare Data

### 2.1 Overview

The varying levels of annotator agreements creates an unusual challenge. The aim is to create a train, test, and validation split that does not bias the model towards obvious sentiments. This guide will describe each step in the data splitting process.

## 2.2 Step 1: Load the Data

Firstly we load the dataset files (which are text files in this case). Considering the possibility that these text files might have different encodings (they initially crashed the program as they use ISO-8859-1), we use Python's `chardet` library to detect the encoding before reading the file. This ensures that the loading of files doesn't fail due to encoding issues. The data is then shuffled randomly. The shuffling was an easy thing to miss, as previously we had used `train_test_split` from `scikit-learn` to split the data, and it had automatically shuffled the data. The data actually comes nearly sorted by sentiment, so without shuffling the model can't be fine-tuned at all.

The files are read in the following way: - Every sentence in the dataset is separated by an '@' delimiter. - There are no headers in the files so they are assigned later as ['sentence', 'label']. - We use pandas' `read_csv` function to read these files.

## 2.3 Step 2: Create a Subset from 'sentences\_50agree'

The `sentences_50agree` dataset is the largest and most diverse dataset. To avoid overfitting and create a diverse validation set, we create a subset from this dataframe that will later be used for validation/testing. This subset is the same size as the train set.

## 2.4 Step 3: Split the Subset into Validation and Test Sets

The subset created from `sentences_50agree` is split into validation and test sets. Here, a stratified sampling is performed to ensure that the split datasets have instances of each class ('positive', 'negative', and 'neutral') in proportions close to their original distribution in the parent dataset. The dataset is split in such way that 31% data is used for validation/testing. We choose 31% as this seems to result in the closest to a 70/15/15 train/validate/test split once the datasets have been assembled.

### Rationale for the 70/15/15 Split:

We specifically chose a 70/15/15 split over other common splits (like 80/10/10) due to class imbalance. The 'Negative' class constitutes only ~12% of the dataset. A smaller test set (e.g., 10%) would result in only ~50 negative examples. This is too small a sample size to reliably evaluate the model's performance (F1 score) on that specific class, as a few misclassifications could drastically swing the metrics. The 15% split ensures we have a statistically significant number of examples (approx. 80) for the minority class in both validation and test sets, providing a more stable evaluation.

## 2.5 Step 4: Update the Training Set

After obtaining the validation and test sets, we remove sentences present in these sets from the training set. This is done to avoid data leakage and ensure that the model doesn't see any sentence in the training phase that it would later encounter in the validation/testing phase.

## 2.6 Step 5: Further Divide the Validation and Test Sets

The validation/test set subset is then divided equally into two parts: one part forms the validation set and the other part forms the test set.

## 2.7 Step 6: Output the Sizes and Percentages

Finally, after creating all the datasets, we output the sizes of the train, validation, and test datasets along with their respective percentages to make sure that the classes are as balanced as they can be given the data.

This completes our data splitting process ensuring that we have a good mix of data in each dataset and our model does not get biased towards the easy-to-identify financial sentiments.

```
[47]: # Load a dataset given its path
def load_data(data_path, seed):
    raw_data = open(data_path, 'rb').read()
    result = chardet.detect(raw_data)
    encoding = result['encoding']
    print("Encoding for:", data_path, "is:", encoding)
    df = pd.read_csv(data_path, delimiter='@', header=None, names=['sentence',
    ↪ 'label'], encoding=encoding)
    return df.sample(frac=1, random_state=seed)

# Load data with correct encoding and delimiter
df_50agree = load_data('Sentences_50Agree.txt', seed_val)
train = load_data(WHICH_DATASET, seed_val)

# Save original count for unit testing purposes
original_50agree_count = train.shape[0]

# Take a random subset of sentences_50agree equal in size to the size of train_
    ↪ dataset
subset_50agree = df_50agree.sample(n=len(train), random_state=seed_val)

# Randomly choose 31% of the data from this subset for validation/test
_, valid_test_subset = train_test_split(subset_50agree, test_size=0.31,
    ↪ random_state=seed_val, stratify=subset_50agree['label'])

# Remove items from train dataset that are in the validation/test set
train = train[~train['sentence'].isin(valid_test_subset['sentence'])]

# Split the validation/test set in half into validation and test set
valid, test = train_test_split(valid_test_subset, test_size=0.5,
    ↪ random_state=seed_val, stratify=valid_test_subset['label'])

# Load pre-trained BERT tokenizer
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')

# Convert texts to BERT input format
train_encodings = tokenizer(train.sentence.tolist(), truncation=True,
    ↪ padding=True)
```



```

val_encodings = tokenizer(valid.sentence.tolist(), truncation=True,
    ↪padding=True)
test_encodings = tokenizer(test.sentence.tolist(), truncation=True,
    ↪padding=True)

def make_tf_dataset(encodings, labels):
    """Function to convert encodings and labels into tensorflow dataset."""
    input_ids = np.array(encodings['input_ids'])
    attention_mask = np.array(encodings['attention_mask'])
    labels = np.array(labels)
    return tf.data.Dataset.from_tensor_slices((input_ids, attention_mask,
    ↪labels)).map(
        lambda input_ids, attention_mask, labels : ({"input_ids": input_ids,
    ↪"attention_mask": attention_mask}, labels))

# Defining the labels keys and their respective values as per the model
label2num = {'negative':0, 'neutral':1, 'positive':2}
train_labels = [label2num[x] for x in train.label.tolist()]
val_labels = [label2num[x] for x in valid.label.tolist()]
test_labels = [label2num[x] for x in test.label.tolist()]

# Convert encodings and labels into tensorflow dataset
train_dataset = make_tf_dataset(train_encodings, train_labels)
val_dataset = make_tf_dataset(val_encodings, val_labels)
test_dataset = make_tf_dataset(test_encodings, test_labels)

# Print the sizes and percentages
total_size = len(train) + len(valid) + len(test)

print("\nSize of Train set:", len(train),
    f"({(len(train)/total_size)*100:.2f}%)")
print("Size of Validation set:", len(valid),
    f"({(len(valid)/total_size)*100:.2f}%)")
print("Size of Test set:", len(test),
    f"({(len(test)/total_size)*100:.2f}%)")

```

Encoding for: Sentences\_50Agree.txt is: ISO-8859-1  
 Encoding for: Sentences\_66Agree.txt is: ISO-8859-1

Size of Train set: 3084 (70.22%)  
 Size of Validation set: 654 (14.89%)  
 Size of Test set: 654 (14.89%)

### 3 Check out the distribution of the classes to ensure they look balanced across datasets

```
[48]: # Distribution in the Training set
print("Training set:")
print(train['label'].value_counts())

# Distribution in the Validation set
print("\nValidation set:")
print(valid['label'].value_counts())

# Distribution in the Test set
print("\nTest set:")
print(test['label'].value_counts())
```

```
Training set:
label
neutral      1854
positive      852
negative      378
Name: count, dtype: int64
```

```
Validation set:
label
neutral      388
positive     186
negative      80
Name: count, dtype: int64
```

```
Test set:
label
neutral      388
positive     185
negative      81
Name: count, dtype: int64
```

### 4 F1 Callback to track F1 Validation

```
[49]: # Custom callback that:
# 1. Computes a weighted F1 score on a held-out validation dataset
# 2. Saves the best model (by F1)
# 3. Logs F1 to Weights & Biases
# 4. Exposes F1 via `logs` so other callbacks (e.g. EarlyStopping) can monitor_
    ↪ it
class F1_Evaluation(tf.keras.callbacks.Callback):
    def __init__(self, validation_data=(), interval=1, save_path=SAVE_PATH):
        """
```

### Parameters

```
validation_data : tf.data.Dataset
    Validation dataset yielding (inputs, labels) batches.
interval : int
    How often to compute F1 (in epochs). interval=1 means every epoch.
save_path : str
    Where to save the best model weights via `model.save_pretrained`.
"""
```

```
super(F1_Evaluation, self).__init__()
```

```
# Track the best F1 score we have seen so far across epochs
```

```
self.best_f1_so_far = 0.0
```

```
self.save_path = save_path
```

```
self.interval = interval
```

```
self.dataset = validation_data
```

```
# We pre-materialize the entire validation set into memory so that:
```

```
# - we do not need to re-iterate over a potentially stateful dataset_
```

→ each epoch

```
# - we can call `model.predict` on plain batches of inputs
```

```
self.X_val, self.y_val = [], []
```

```
# Iterate once over the validation dataset to collect input and label_
```

→ batches

```
for batch in self.dataset:
```

```
    # Each batch is a tuple of (inputs, labels)
```

```
    inputs, labels = batch
```

```
    # `inputs` is a dict with keys like ["input_ids", "attention_mask"]
```

```
    self.X_val.append(inputs)
```

```
    # Convert label tensor to a 1D NumPy array, even if there is a_
```

→ single label

```
    # 1. `labels.numpy()` -> NumPy
```

```
    # 2. `np.squeeze` removes singleton dimensions
```

```
    # 3. `np.atleast_1d` guarantees it is at least 1D
```

```
    self.y_val.append(np.atleast_1d(np.squeeze(labels.numpy())))
```

```
# Keep a history of F1 scores across epochs (for later analysis /_
```

→ plotting)

```
self.f1_scores = []
```

```
def on_epoch_end(self, epoch, logs=None):
```

```
    """
```

```
    Called by Keras at the end of each epoch.
```

```
    We compute F1 on the pre-saved validation set and:
```

```

- update `self.best_f1_so_far` and save the model if it improved
- log F1 to W&B
- inject `val_f1_score` into `logs` so that EarlyStopping can monitor it
"""

# Always work with a dict (Keras passes one in, but we keep this
→defensive)
if logs is None:
    logs = {}

print("\nCalculating validation F1 score...")

# Only compute F1 on epochs that match the specified interval
# (e.g., every epoch if interval=1, every 2 epochs if interval=2, etc.)
if epoch % self.interval == 0:
    y_pred = []

    # Loop over the cached validation batches and get predictions
    for X_val_batch in self.X_val:
        # Suppress verbose output from `model.predict` for cleaner logs
        with contextlib.redirect_stdout(io.StringIO()), \
            contextlib.redirect_stderr(io.StringIO()):
            # For a Hugging Face TF model, `.predict` returns an object
            # with a `.logits` attribute
            y_pred_single = self.model.predict(X_val_batch).logits

            # Convert logits to probabilities with softmax, then take argmax
            # to get the predicted class index for each example
            y_pred_single = tf.nn.softmax(y_pred_single).numpy().
→argmax(axis=-1)

            # Again, make sure each batch prediction is at least 1D before
→stacking
            y_pred.append(np.atleast_1d(y_pred_single))

    # Concatenate all batch predictions and labels into flat arrays
    y_pred = np.concatenate(y_pred)
    y_val = np.concatenate(self.y_val)

    # Compute weighted F1 score to account for possible class imbalance
    _f1 = f1_score(y_val, y_pred, average='weighted')

    # If this epoch's F1 is the best so far, update our record and save
→the model
    if _f1 > self.best_f1_so_far:
        self.best_f1_so_far = _f1
        print("Found new best F1 score:", _f1, "Saving the model.")
        # For HF models, `save_pretrained` saves the weights + config

```

```

        self.model.save_pretrained(self.save_path)

        # Store F1 in the callback's internal history
        self.f1_scores.append(_f1)

        # Human-readable printout to the notebook
        print("Epoch: {} - validation_data f1_score: {:.4f}".format(epoch + 1, _f1))

        # Make F1 available to other callbacks (e.g., EarlyStopping) via `logs`.
        logs["val_f1_score"] = _f1

        # EarlyStopping can now use `monitor="val_f1_score", mode="max".

        # Log to Weights & Biases so we can track F1 over time in the UI
        wandb.log({"val_f1_score": _f1})

```

```

[50]: # Log in to W&B account for experiment tracking
wandb.login()

# Load pre-trained BERT model with a classification layer on top
# Added use_safetensors=False to address TypeError: 'builtins.safe_open' object
# is not iterable
model = TFBertForSequenceClassification.from_pretrained('bert-base-uncased',
num_labels=3, use_safetensors=False)

# Define loss function, optimizer, and metrics
loss_fn = keras.losses.SparseCategoricalCrossentropy(from_logits=True)

# Learning-rate schedule: start with a small LR and decay it exponentially over
# time.
lr_schedule = keras.optimizers.schedules.ExponentialDecay(
    initial_learning_rate=3e-5, # initial LR (good start for BERT fine-tuning)
    decay_steps=10000,          # after this many "steps" (batches), apply one
    decay                        # decay
    decay_rate=0.9              # multiply the LR by 0.9 every `decay_steps`
)

# Adam optimizer that uses the schedule above instead of a fixed learning rate.
optimizer = keras.optimizers.Adam(learning_rate=lr_schedule)

# Compile the model (tell Keras how to train it)
model.compile(
    optimizer=optimizer, # optimization algorithm (Adam with the LR schedule)
    loss=loss_fn,        # loss function to minimize (sparse categorical
    crossentropy)

```

```

    metrics=['accuracy'] # metrics to report during training/validation
    ↪(classification accuracy)
)

# Initializing wandb run
wandb.init(project="bert-sentiment", name=WHICH_RUN_NAME)

# Prepare for F1 score evaluation
val_dataset_temp = val_dataset.batch(8)
f1_evaluation = F1_Evaluation(validation_data=val_dataset_temp, interval=1,
    ↪save_path=SAVE_PATH)

# Add early stopping (disabled for now)
early_stopping = EarlyStopping(monitor='val_f1_score', patience=10,
    ↪restore_best_weights=True)

# Train the model
history = model.fit(
    train_dataset.batch(8),
    validation_data=val_dataset_temp,
    epochs=30,
    callbacks=[WandbCallback(save_model=False), f1_evaluation, early_stopping]
)

```

wandb: **WARNING** Calling wandb.login() after wandb.init() has no effect.  
All model checkpoint layers were used when initializing TFBertForSequenceClassification.

Some layers of TFBertForSequenceClassification were not initialized from the model checkpoint at bert-base-uncased and are newly initialized: ['classifier']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

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```

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Epoch 1/30

386/386 [=====] - ETA: 0s - loss: 0.5081 - accuracy: 0.8035

Calculating validation F1 score...

Found new best F1 score: 0.834046247940936 Saving the model.

Epoch: 1 - validation\_data f1\_score: 0.8340

386/386 [=====] - 111s 153ms/step - loss: 0.5081 - accuracy: 0.8035 - val\_loss: 0.4376 - val\_accuracy: 0.8349 - val\_f1\_score: 0.8340

Epoch 2/30

386/386 [=====] - ETA: 0s - loss: 0.2152 - accuracy: 0.9283

Calculating validation F1 score...

Epoch: 2 - validation\_data f1\_score: 0.8134

386/386 [=====] - 33s 86ms/step - loss: 0.2152 - accuracy: 0.9283 - val\_loss: 0.6399 - val\_accuracy: 0.8104 - val\_f1\_score: 0.8134

Epoch 3/30

386/386 [=====] - ETA: 0s - loss: 0.1124 - accuracy: 0.9663

Calculating validation F1 score...

Epoch: 3 - validation\_data f1\_score: 0.8322

386/386 [=====] - 32s 83ms/step - loss: 0.1124 - accuracy: 0.9663 - val\_loss: 0.6115 - val\_accuracy: 0.8318 - val\_f1\_score: 0.8322

Epoch 4/30

386/386 [=====] - ETA: 0s - loss: 0.0760 - accuracy: 0.9776

Calculating validation F1 score...

Found new best F1 score: 0.8377560893829558 Saving the model.

Epoch: 4 - validation\_data f1\_score: 0.8378

386/386 [=====] - 36s 93ms/step - loss: 0.0760 - accuracy: 0.9776 - val\_loss: 0.6877 - val\_accuracy: 0.8379 - val\_f1\_score: 0.8378

Epoch 5/30

386/386 [=====] - ETA: 0s - loss: 0.0573 - accuracy: 0.9802

Calculating validation F1 score...

Epoch: 5 - validation\_data f1\_score: 0.8118

386/386 [=====] - 33s 85ms/step - loss: 0.0573 - accuracy: 0.9802 - val\_loss: 0.7497 - val\_accuracy: 0.8150 - val\_f1\_score: 0.8118

Epoch 6/30

386/386 [=====] - ETA: 0s - loss: 0.0334 - accuracy: 0.9890

Calculating validation F1 score...

Epoch: 6 - validation\_data f1\_score: 0.8165

386/386 [=====] - 32s 82ms/step - loss: 0.0334 - accuracy: 0.9890 - val\_loss: 0.7357 - val\_accuracy: 0.8150 - val\_f1\_score: 0.8165

Epoch 7/30

386/386 [=====] - ETA: 0s - loss: 0.0188 - accuracy: 0.9938

Calculating validation F1 score...

Epoch: 7 - validation\_data f1\_score: 0.8276

386/386 [=====] - 32s 82ms/step - loss: 0.0188 - accuracy: 0.9938 - val\_loss: 0.9351 - val\_accuracy: 0.8272 - val\_f1\_score: 0.8276

Epoch 8/30

386/386 [=====] - ETA: 0s - loss: 0.0187 - accuracy: 0.9951

Calculating validation F1 score...

Epoch: 8 - validation\_data f1\_score: 0.8106

386/386 [=====] - 32s 83ms/step - loss: 0.0187 - accuracy: 0.9951 - val\_loss: 1.0235 - val\_accuracy: 0.8073 - val\_f1\_score: 0.8106

Epoch 9/30

386/386 [=====] - ETA: 0s - loss: 0.0452 - accuracy: 0.9864

Calculating validation F1 score...

Epoch: 9 - validation\_data f1\_score: 0.8269

386/386 [=====] - 32s 82ms/step - loss: 0.0452 - accuracy: 0.9864 - val\_loss: 0.7343 - val\_accuracy: 0.8272 - val\_f1\_score: 0.8269

Epoch 10/30

386/386 [=====] - ETA: 0s - loss: 0.0242 - accuracy: 0.9938

Calculating validation F1 score...

Epoch: 10 - validation\_data f1\_score: 0.8230

386/386 [=====] - 32s 82ms/step - loss: 0.0242 - accuracy: 0.9938 - val\_loss: 0.9099 - val\_accuracy: 0.8242 - val\_f1\_score: 0.8230

Epoch 11/30

386/386 [=====] - ETA: 0s - loss: 0.0218 - accuracy: 0.9925

Calculating validation F1 score...

Epoch: 11 - validation\_data f1\_score: 0.8359

386/386 [=====] - 32s 82ms/step - loss: 0.0218 - accuracy: 0.9925 - val\_loss: 0.8780 - val\_accuracy: 0.8364 - val\_f1\_score: 0.8359

Epoch 12/30

386/386 [=====] - ETA: 0s - loss: 0.0132 - accuracy: 0.9974



Calculating validation F1 score...

Epoch: 12 - validation\_data f1\_score: 0.8269

386/386 [=====] - 32s 82ms/step - loss: 0.0132 - accuracy: 0.9974 - val\_loss: 1.0217 - val\_accuracy: 0.8287 - val\_f1\_score: 0.8269

Epoch 13/30

386/386 [=====] - ETA: 0s - loss: 0.0494 - accuracy: 0.9835

Calculating validation F1 score...

Epoch: 13 - validation\_data f1\_score: 0.8243

386/386 [=====] - 32s 82ms/step - loss: 0.0494 - accuracy: 0.9835 - val\_loss: 0.8887 - val\_accuracy: 0.8242 - val\_f1\_score: 0.8243

Epoch 14/30

386/386 [=====] - ETA: 0s - loss: 0.0274 - accuracy: 0.9925

Calculating validation F1 score...

Epoch: 14 - validation\_data f1\_score: 0.8241

386/386 [=====] - 32s 82ms/step - loss: 0.0274 - accuracy: 0.9925 - val\_loss: 1.0843 - val\_accuracy: 0.8226 - val\_f1\_score: 0.8241

Epoch 15/30

386/386 [=====] - ETA: 0s - loss: 0.0122 - accuracy: 0.9968

Calculating validation F1 score...

Found new best F1 score: 0.838021713917435 Saving the model.

Epoch: 15 - validation\_data f1\_score: 0.8380

386/386 [=====] - 36s 94ms/step - loss: 0.0122 - accuracy: 0.9968 - val\_loss: 1.0496 - val\_accuracy: 0.8394 - val\_f1\_score: 0.8380

Epoch 16/30

386/386 [=====] - ETA: 0s - loss: 0.0117 - accuracy: 0.9964

Calculating validation F1 score...

Found new best F1 score: 0.8423858033290371 Saving the model.

Epoch: 16 - validation\_data f1\_score: 0.8424

386/386 [=====] - 36s 94ms/step - loss: 0.0117 - accuracy: 0.9964 - val\_loss: 0.9273 - val\_accuracy: 0.8425 - val\_f1\_score: 0.8424

Epoch 17/30

386/386 [=====] - ETA: 0s - loss: 0.0223 - accuracy: 0.9945

Calculating validation F1 score...

Epoch: 17 - validation\_data f1\_score: 0.8423

386/386 [=====] - 32s 84ms/step - loss: 0.0223 - accuracy: 0.9945 - val\_loss: 0.9441 - val\_accuracy: 0.8456 - val\_f1\_score: 0.8423

Epoch 18/30

```

386/386 [=====] - ETA: 0s - loss: 0.0262 - accuracy:
0.9929
Calculating validation F1 score...
Epoch: 18 - validation_data f1_score: 0.8289
386/386 [=====] - 32s 82ms/step - loss: 0.0262 -
accuracy: 0.9929 - val_loss: 1.0631 - val_accuracy: 0.8287 - val_f1_score:
0.8289

```

## 5 Plot the training

```

[51]: def plot_training_process(history, f1_scores):
    fig, axs = plt.subplots(2, 2, figsize=(18, 8),
                           gridspec_kw={'height_ratios': [2, 1]})

    # Collapse the subplots for the F1 scores into a single subplot
    axs[1, 0].remove()
    axs[1, 1].remove()

    # Plotting training and validation loss
    axs[0, 0].plot(history.history["loss"], label='Train')
    axs[0, 0].plot(history.history["val_loss"], label='Validation')
    axs[0, 0].set_title("Model Loss")
    axs[0, 0].set_ylabel("Loss")
    axs[0, 0].set_xlabel("Epoch")
    axs[0, 0].legend()

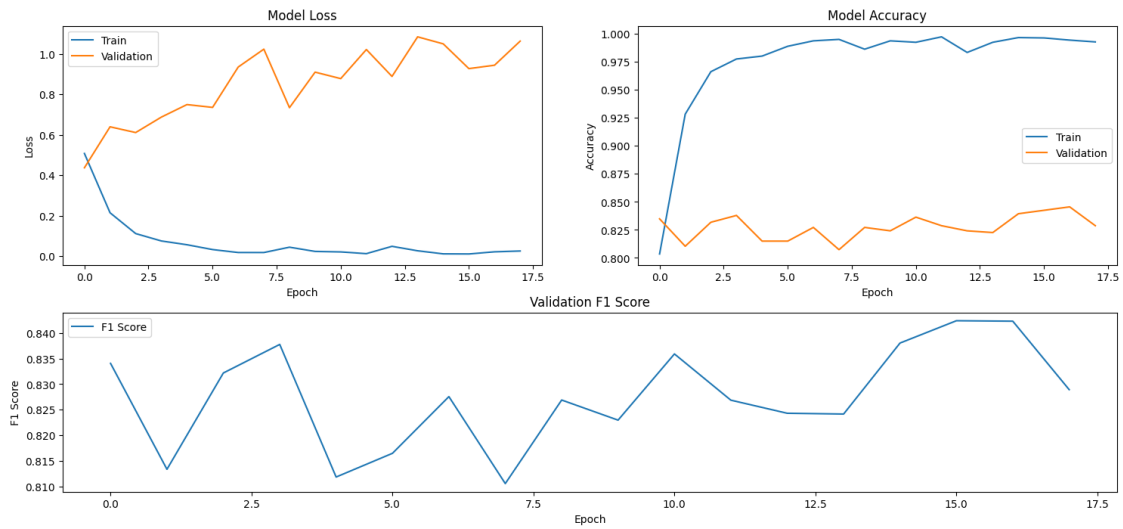
    # Plotting training and validation accuracy
    axs[0, 1].plot(history.history["accuracy"], label='Train')
    if "val_accuracy" in history.history:
        axs[0, 1].plot(history.history["val_accuracy"], label='Validation')
    axs[0, 1].set_title("Model Accuracy")
    axs[0, 1].set_ylabel("Accuracy")
    axs[0, 1].set_xlabel("Epoch")
    axs[0, 1].legend()

    # Plotting Validation F1 Scores
    ax2 = fig.add_subplot(2, 1, 2) # 2x1, occupying the second slot.
    ax2.plot(f1_scores)
    ax2.set_title("Validation F1 Score")
    ax2.set_ylabel("F1 Score")
    ax2.set_xlabel("Epoch")
    ax2.legend(["F1 Score"], loc="upper left")

    # Improve layout and show plots
    plt.subplots_adjust(hspace=0.6)
    plt.show()

```

```
[52]: plot_training_process(history, f1_evaluation.f1_scores)
```



## 6 Evaluate with the test data on the checkpointed model

```
[53]: # Load the checkpointed model
finetuned_model = TFBertForSequenceClassification.from_pretrained(SAVE_PATH)
finetuned_model.trainable = False

# Evaluate the model with test dataset and show the classification report
y_test = test.label.map(label2num).values
y_pred = tf.nn.softmax(finetuned_model.predict(test_dataset.batch(8)).logits).
    ↪ numpy().argmax(axis=-1)

print(classification_report(y_test, y_pred, target_names=['negative',
    ↪ 'neutral', 'positive']))
test_f1_score = f1_score(y_test, y_pred, average='weighted')
print("Test F1 Score:", test_f1_score)

# Log the test F1 score to wandb (single data point)
wandb.log({'test_f1_score': test_f1_score})
```

Some layers from the model checkpoint at /content/drive/My Drive/finetuned\_bert\_66\_30 were not used when initializing TFBertForSequenceClassification: ['dropout\_265']

- This IS expected if you are initializing TFBertForSequenceClassification from the checkpoint of a model trained on another task or with another architecture (e.g. initializing a BertForSequenceClassification model from a BertForPreTraining model).

- This IS NOT expected if you are initializing TFBertForSequenceClassification from the checkpoint of a model that you expect to be exactly identical (initializing a BertForSequenceClassification model from a BertForSequenceClassification model).

All the layers of TFBertForSequenceClassification were initialized from the model checkpoint at /content/drive/My Drive/finetuned\_bert\_66\_30.

If your task is similar to the task the model of the checkpoint was trained on, you can already use TFBertForSequenceClassification for predictions without further training.

82/82 [=====] - 5s 24ms/step

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.84      | 0.80   | 0.82     | 81      |
| neutral      | 0.88      | 0.85   | 0.87     | 388     |
| positive     | 0.73      | 0.80   | 0.76     | 185     |
| accuracy     |           |        | 0.83     | 654     |
| macro avg    | 0.82      | 0.82   | 0.82     | 654     |
| weighted avg | 0.84      | 0.83   | 0.83     | 654     |

Test F1 Score: 0.8330028413355849